Aus Wildfire - Withheld-Seasons

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With the following RMSE difference figures, we look to compare the difference in RMSE between the All-data models and withheld-season models (both the prediction season and another season), given in Figures 1, 2, and 3. Or we compare the difference in RMSE between withholding the prediction season and withheld-season models (including other seasons), in Figures 4, 5, and 6.

While these are interesting figures, I believe that we are looking to explore two questions. First, is there a change over time of the underlying climate-CO connections where the inclusion/exclusion of a season would affect the predictions (using RMSE)? Second, does the inclusion of a single strong wildfire season affect the model and predictions of other seasons? In both cases the single withheld-season Figures (4, 5, and 6) allow for easier interpretation.

For the first question, if we observe a change in the underlying relationship then the further away, in time, a wildfire-season was then its removal should improve (reduce) the RMSE of the prediction year. This would show up as an increase in ΔRMSE in as we move away from the diagonal along the x-axis in Figures 4, 5, and 6. However, I do not observe this in any of the groups (fire-seasons).

For the second question, we would observe increases in ΔRMSE in vertical bands above particularly strong wildfire seasons, such as 2019/2020. Again, I do not observe this in any figures, including Figures 4, 5, and 6.

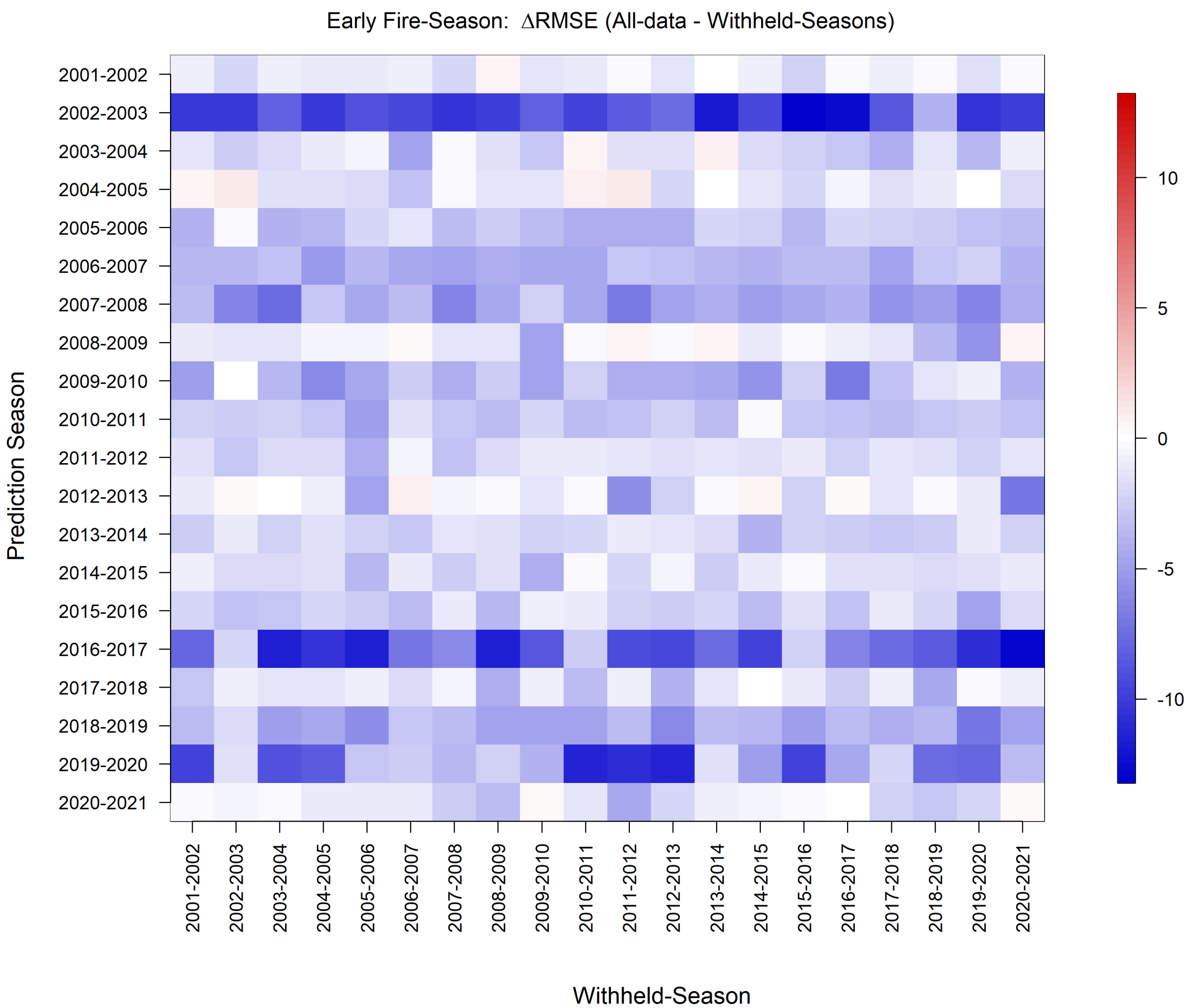

Early Fire-Season: ΔRMSE (All-data - Withheld-Seasons)
Prediction Season
2001-2002
2002-2003
2003-2004
2004-2005
2005-2006
2006-2007
2007-2008
2008-2009
2009-2010
2010-2011
2011-2012
2012-2013
2013-2014
2014-2015
2015-2016
2016-2017
2017-2018
2018-2019
2019-2020
2020-2021
Withheld-Season
10
5
0
-5
-10

Figure 1

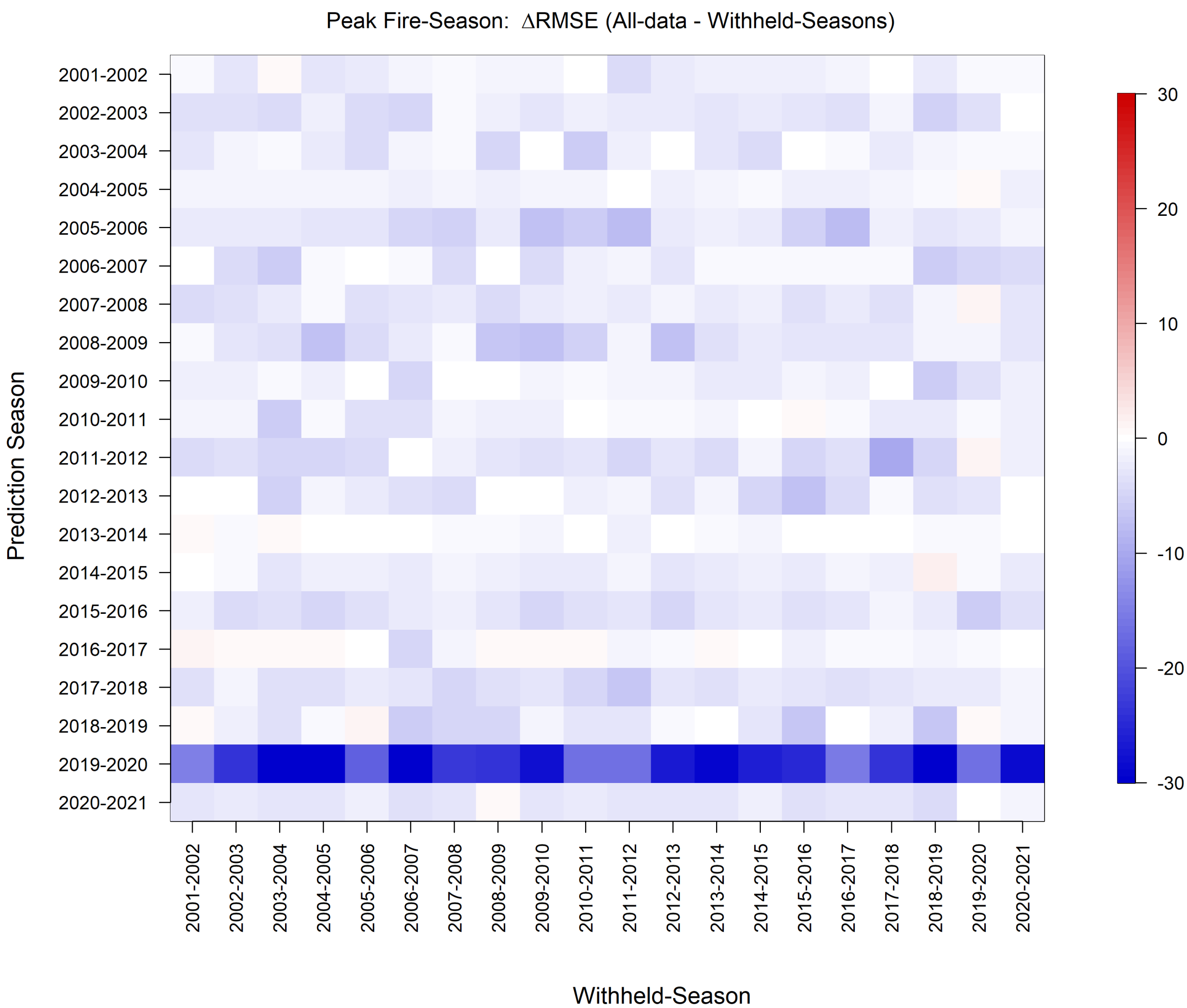

Peak Fire-Season: ΔRMSE (All-data - Withheld-Seasons)
Prediction Season
Withheld-Season
2001-2002
2002-2003
2003-2004
2004-2005
2005-2006
2006-2007
2007-2008
2008-2009
2009-2010
2010-2011
2011-2012
2012-2013
2013-2014
2014-2015
2015-2016
2016-2017
2017-2018
2018-2019
2019-2020
2020-2021
30
20
10
0
-10
-20
-30

Figure 2

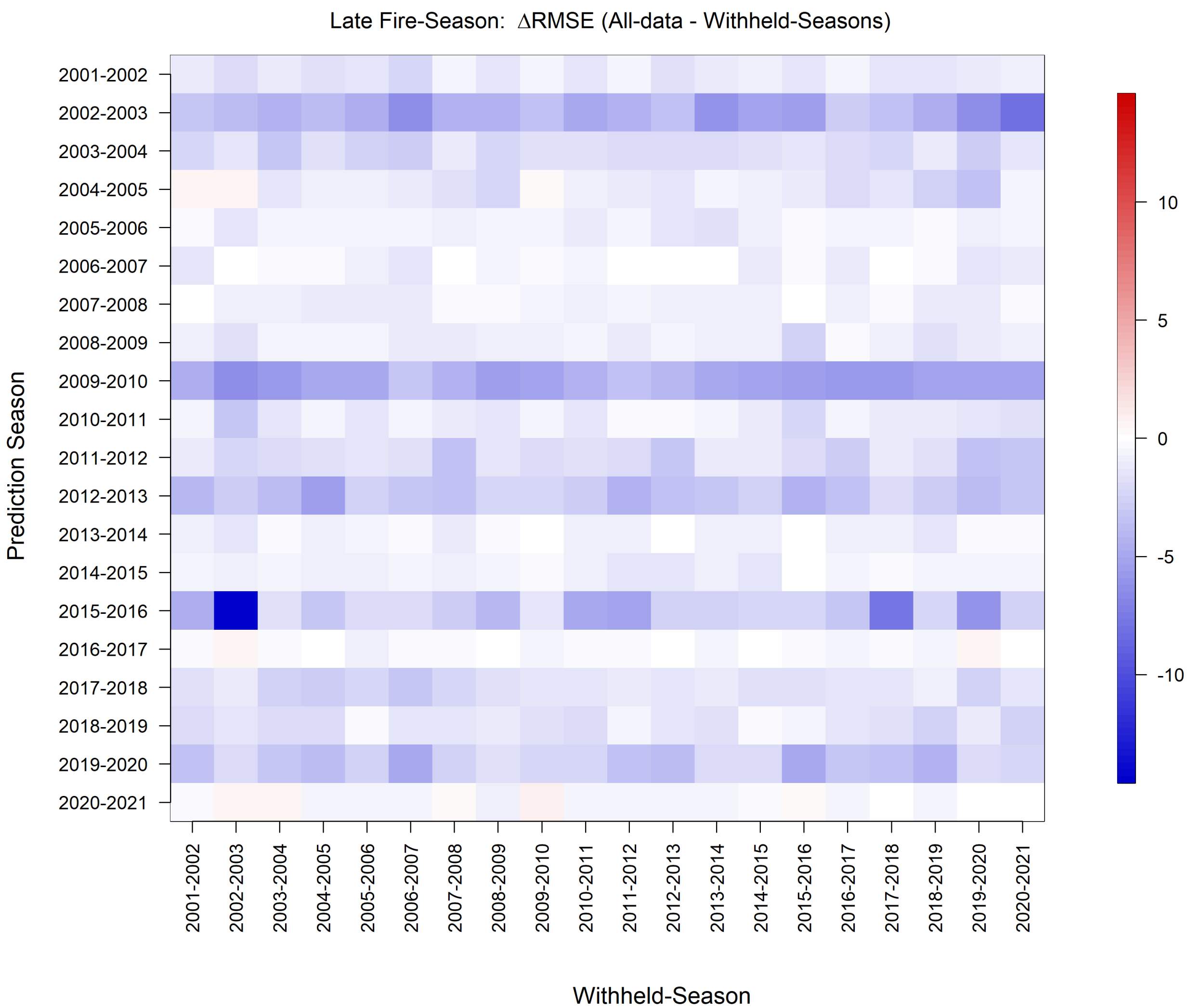
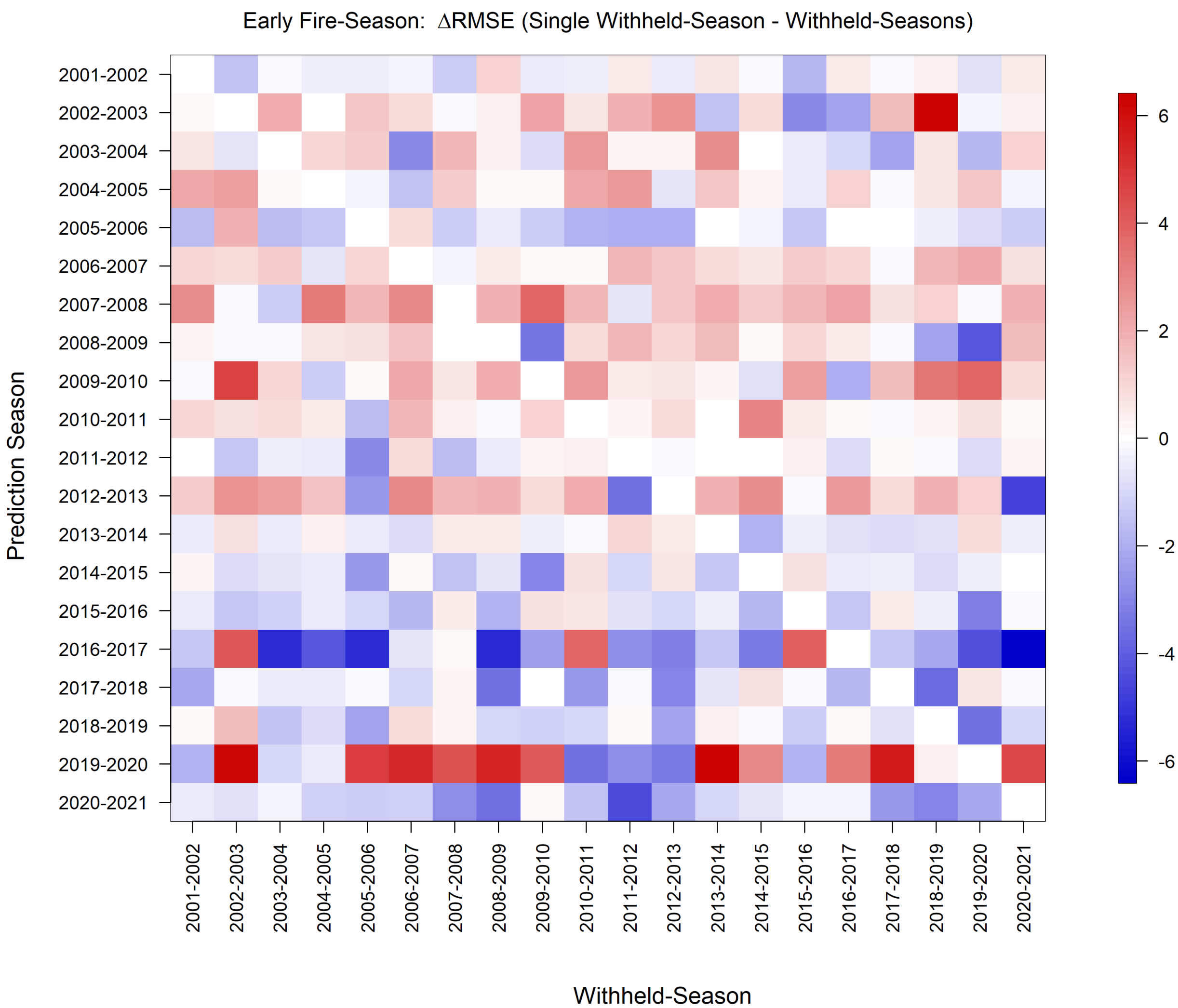

Late Fire-Season: ΔRMSE (All-data - Withheld-Seasons)
Prediction Season
2001-2002
2002-2003
2003-2004
2004-2005
2005-2006
2006-2007
2007-2008
2008-2009
2009-2010
2010-2011
2011-2012
2012-2013
2013-2014
2014-2015
2015-2016
2016-2017
2017-2018
2018-2019
2019-2020
2020-2021
Withheld-Season
10
5
0
-5
-10

Figure 3



Early Fire-Season: ΔRMSE (Single Withheld-Season - Withheld-Seasons)
Prediction Season
2001-2002
2002-2003
2003-2004
2004-2005
2005-2006
2006-2007
2007-2008
2008-2009
2009-2010
2010-2011
2011-2012
2012-2013
2013-2014
2014-2015
2015-2016
2016-2017
2017-2018
2018-2019
2019-2020
2020-2021
Withheld-Season
6
4
2
0
-2
-4
-6

Figure 4

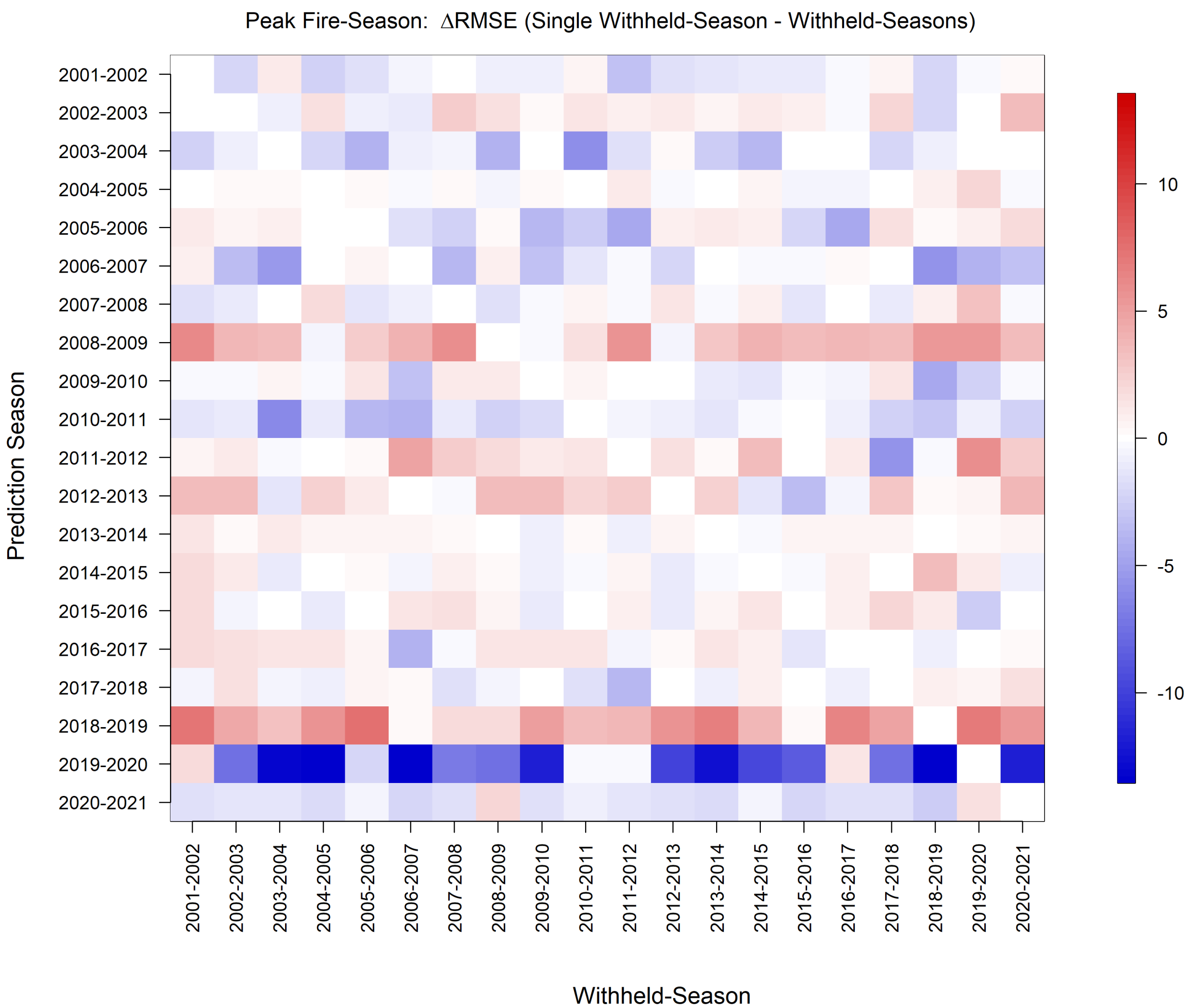

Peak Fire-Season: ΔRMSE (Single Withheld-Season - Withheld-Seasons)
Prediction Season
2001-2002
2002-2003
2003-2004
2004-2005
2005-2006
2006-2007
2007-2008
2008-2009
2009-2010
2010-2011
2011-2012
2012-2013
2013-2014
2014-2015
2015-2016
2016-2017
2017-2018
2018-2019
2019-2020
2020-2021
Withheld-Season
10
5
0
-5
-10

Figure 5

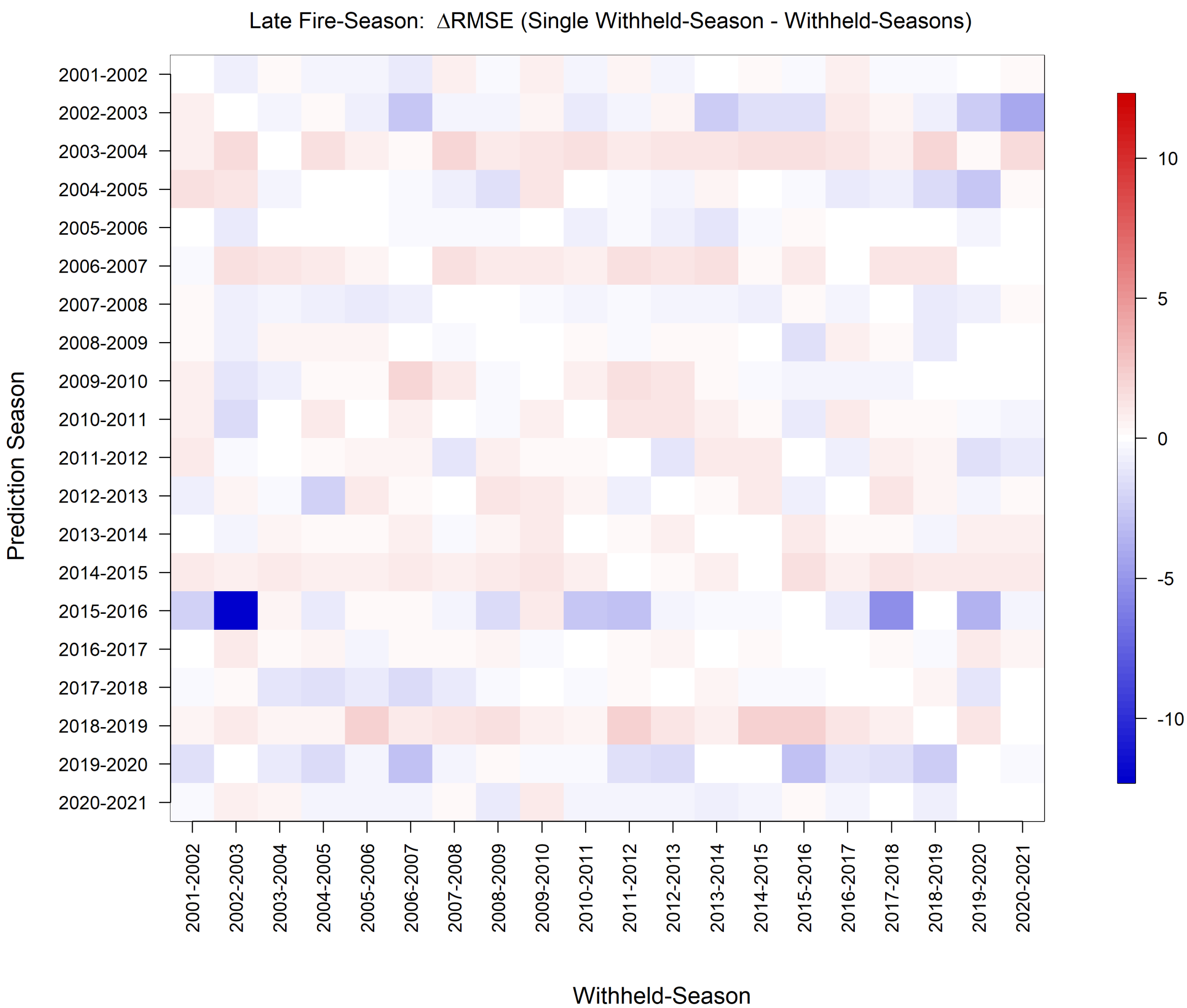

Late Fire-Season: ΔRMSE (Single Withheld-Season - Withheld-Seasons)
Prediction Season
2001-2002
2002-2003
2003-2004
2004-2005
2005-2006
2006-2007
2007-2008
2008-2009
2009-2010
2010-2011
2011-2012
2012-2013
2013-2014
2014-2015
2015-2016
2016-2017
2017-2018
2018-2019
2019-2020
2020-2021
Withheld-Season
10
5
0
-5
-10

Figure 6